

Understanding the

Maintenance Analysis Tab

Summary Edition — A Condensed Learner's Guide

Tab 9 of app_jao_NP_API_fix_d14.py
Nordic Flow-Based Propagation — fi_no3

Every chapter and formula below is condensed from the full 56-chapter edition — same numbering, same content, abridged to its opening idea and central formula.

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Preface

This book teaches the Maintenance Analysis tab from the ground up, including the actual mathematics underneath every button — with particular depth on the hypothesis-testing machinery, since that is what "Run Analysis" and every verdict downstream of it actually rests on. It does not assume you already know what a fixed-effects estimator is, how a clustered standard error is computed, why a p-value means what it means, or what a gradient-boosted tree actually optimizes. Every idea is derived, not merely cited, and the estimation and hypothesis-testing chapters include fully worked numeric examples — real numbers, computed by hand, that you can reproduce yourself.

The tab itself is one of nine tabs in an extended tkinter application (`app_jao_NP_API_fix_d14.py`) built on a shared analytical backend (`propagation.py`). It re-runs, inside one self-contained corner of that application, the same validation the rest of the project performs: does a Nordic country's maintenance outage measurably move another bidding zone's published flow-based market parameters?

How this book is organised. Part I builds the domain vocabulary and the one governing accounting formula. Part II builds the full statistical machinery from scratch — the population regression model, the OLS derivation, the Gauss–Markov theorem, panel fixed effects and the omitted-variable-bias problem they solve, heteroskedasticity and clustered standard errors, confidence intervals, Type I/Type II error and power, and multiple-testing correction — each with a derivation and at least one fully worked numeric example. Part III is the mathematical centre of the book: each of the six hypotheses this tab tests, H1 through H6, given its own chapter — formal null and alternative hypotheses, full regression specification, worked numeric interpretation, and the specific pitfalls each one guards against — followed by chapters on economic significance, collinearity, and one complete walkthrough from a raw outage event to a final verdict. Part IV introduces counterfactual (interrupted-time-series) modeling and the one rule that governs it. Part V derives every one of the fifteen counterfactual methods. Part VI tours the tab itself, sub-tab by sub-tab. Part VII is for whoever maintains this code next. Three appendices collect notation, a formula quick-reference, and a glossary.

Notation used throughout: subscript i indexes a CNEC (a cross-sectional entity); subscript t indexes a timestamp; subscript g indexes a cluster; a hat (e.g. $\hat{\beta}$) denotes an estimated quantity; a bar (e.g. $\bar{\beta}$) denotes an average; a tilde (e.g. $\tilde{\beta}$) denotes a

within-transformed (demeaned) quantity. Read Parts I–V once, in order, with a pencil — work the examples yourself before reading the stated answer. After that, this book works equally well as a reference.

Part I — Foundations

Chapter 1. What This Tab Does, and Why It Exists

Nordic transmission system operators (TSOs) — Fingrid in Finland, Svenska kraftnät in Sweden, Statnett in Norway, Energinet in Denmark — take planned and forced maintenance outages on their own grids.

Chapter 2. The Vocabulary You Need

FBMC replaces the older, more intuitive idea of trading capacity zone-pair by zone-pair with a different question: for every critical line in the grid, how much can every zone's net position change before that specific line's limit is reached? The market-clearing algorithm then finds the prices and volumes that respect every critical line's limit simultaneously.

$$RAM = Fmax - FRM - fall + fnrao + AMR - FAAC - IVA$$

Chapter 3. The RAM Identity — The Formula Underneath Everything

The first formula above is an accounting identity, not an approximation. Every term is independently published in the market data, and their signed sum equals the published RAM column to within about one megawatt on real data.

$$RAM_t = \sum_{k=1}^7 s_k \cdot X_{k,t}, \text{ with } s = (+1, -1, -1, +1, +1, -1, -1)$$

Part II — The Mathematics of Estimation

This part builds, from nothing, every piece of statistical machinery Part III leans on. Nothing here is specific to power grids — it is the general theory of estimating a treatment effect from observational panel data, worked with actual numbers so each formula has a face.

Chapter 4. The Population Regression Function and the Assumptions Behind It

Start from the question itself: does some outage-related regressor x cause a change in some outcome y ? The **population regression function** posits a true, unobserved relationship

$$y_i = \alpha + \beta x_i + \varepsilon_i$$

Chapter 5. Deriving the OLS Estimator

Given a sample of n observations (x_i, y_i) , **Ordinary Least Squares** chooses the values α , β that make the fitted line come as close as possible to the data, in the specific sense of minimising the sum of squared vertical distances (residuals):

$$S(\alpha, \beta) = \sum_{i=1}^n (y_i - \alpha - \beta x_i)^2$$

Chapter 6. A Fully Worked OLS Example, By Hand

Six observations of a RAM-like outcome, three before an outage and three during it:

$$\beta = \text{Cov}(x, y) / \text{Var}(x) = -5.333, \alpha = \bar{y} - \beta \bar{x} = 100.333$$

Chapter 7. The Gauss–Markov Theorem: Why OLS Is BLUE

Why square the errors at all, rather than some other penalty?

Chapter 8. Goodness of Fit: R^2 , Within R^2 , and Overall R^2

R^2 measures the fraction of y 's variation the fitted model explains:

$$R^2 = 1 - \text{SSE} / \text{SST}, \text{SST} = \sum (y_i - \bar{y})^2$$

Chapter 9. Panel Data and the Fixed-Effects (Within) Estimator

This tab's data isn't one cross-section — it's a **panel**: many CNECs (indexed i), each observed repeatedly over time (indexed t). Pooling every (i,t) row into one plain OLS regression like Chapter 5's implicitly assumes every CNEC shares the same baseline level, which is false, and Chapter 10 shows concretely how wrong ignoring that can be.

$$y_{it} = \alpha_i + \beta D_{it} + \varepsilon_{it}$$

Chapter 10. A Fully Worked Fixed-Effects Example: Omitted-Variable Bias, Made Concrete

Two CNECs, deliberately unbalanced: CNEC A (high baseline RAM) is observed mostly **before** an outage; CNEC B (low baseline RAM) is observed mostly **during** one.

$$A: \bar{y}_A = 197.8 \quad B: \bar{y}_B = 146.0$$

Chapter 11. The Frisch–Waugh–Lovell Theorem: What “Controlling For” Really Means

Chapter 9 asserted, without proof, that demeaning is equivalent to including a full set of entity dummies. The **Frisch–Waugh–Lovell (FWL) theorem** is the general statement behind that claim, and behind the whole idea of “controlling for” a variable in any regression.

Chapter 12. Heteroskedasticity and the Robust (White) Standard Error

Gauss–Markov's variance formula, $\sigma^2(X'X)^{-1}$, assumed every residual has the **same** variance σ^2 . **Heteroskedasticity** — unequal error variance across observations — is the norm here, not the exception: a CNEC that rarely binds has shadow price pinned near zero with little variance; one that binds often has much more.

$$\text{Var}_{\text{robust}}(\beta) = (X'X)^{-1} \left[\sum_i u_i^2 x_i x_i' \right] (X'X)^{-1}$$

Chapter 13. Clustered Standard Errors, Derived and Worked

Observations from the same time period are not independent: a shock at 14:00 on one date plausibly affects several CNECs simultaneously (shared weather, a shared market

condition), so errors from *different* CNECs at the *same timestamp* are correlated, even though errors from *different timestamps* are more nearly independent.

$$\text{Var}_{\text{cluster}}(\beta) = (X^T X)^{-1} \left[\sum_{g=1}^G X_g^T u_g u_g^T X_g \right] (X^T X)^{-1}$$

Chapter 14. Confidence Intervals and the Duality With Hypothesis Tests

A point estimate β alone hides how much sampling variability it carries. A **confidence interval** expresses that variability directly, as a range of values plausible given the data:

$$\beta \pm t_{\alpha/2, df} \cdot SE(\beta)$$

Chapter 15. Type I and Type II Error, Power, and Sample Size

Every hypothesis test can go wrong in exactly two ways, and the two are in tension with each other:

Chapter 16. Multiple Testing: Family-Wise Error Rate and the Holm–Bonferroni Procedure, Worked

Run one test at $\alpha=0.05$ and the false-positive rate is, by construction, 5%. Run m *independent* tests and ask for the probability that *at least one* falsely rejects — the **family-wise error rate (FWER)** — and it climbs:

$$\text{FWER} = 1 - (1 - \alpha)^m$$

Part III — The Six Hypotheses, In Full

Every regression below is an instance of Chapter 9's fixed-effects model, fit with Chapter 13's clustered standard errors, read through Chapters 14–16's testing framework. This part gives each hypothesis its own chapter: the formal null and alternative, the exact specification, what a verdict means substantively, and the pitfall it specifically guards against.

Chapter 17. H1 in Full: HVDC Outages and the Reference Flow

`fall` is the sign-normalised reference flow (“`fall_signed`,” falling back to raw `fall` if unavailable), constructed so a source-country HVDC outage's effect has a *consistent expected sign* regardless of any individual CNEC's own flow-direction convention — without this normalisation, genuinely identical physical effects on differently-oriented CNECs could cancel each other out in the pooled panel rather than reinforcing.

$$H_0: \beta = 0 \text{ vs. } H_A: \beta \neq 0$$

Chapter 18. H2 in Full: AC Line Outages and PTDF Sensitivity

Using the **absolute** PTDF value is the specification choice that makes this a clean test of a pure topology effect: an AC line outage changes the grid's admittance matrix, which changes how *sensitive* a CNEC is to the source zone's net position — the sensitivity can grow or shrink depending on which lines are still available, but $|\text{PTDF}|$ growing either way is the signature of a topology change, independent of how much power happens to be flowing at any given moment (which is what H1's flow-level test already covers).

$$H_0: \beta = 0 \text{ vs. } H_A: \beta \neq 0$$

Chapter 19. H3 in Full: Forced Outages and RAM

Because RAM is itself Chapter 3's seven-term identity, a significant β says the *combined*, already-netted effect across every term moved — it does not by itself say which term(s) moved or by how much.

$$H_0: \beta = 0 \text{ vs. } H_A: \beta \neq 0$$

Chapter 20. H4 in Full: Forced Outages and the Shadow Price

H4 is fit on the **binding subsample** only — rows where the CNEC actually has a positive shadow price — since a non-binding CNEC's shadow price is exactly zero *by construction* (Chapter 2), not because nothing happened; including those structural zeros would mix two different questions, "did the outage make this constraint bind harder" and "was this CNEC ever going to bind regardless," into one coefficient that answers neither cleanly.

$$H_{\blacksquare}: \beta_{\blacksquare} = 0 \text{ vs. } H_{\blacksquare}: \beta_{\blacksquare} \neq 0$$

Chapter 21. H5 in Full: The Logistic Regression for IVA, Derived and Worked

IVA is a discretionary correction, usually exactly zero, so H5 cannot be a linear regression on IVA's *size* the way H1–H4 are on their outcomes — it is a test of whether IVA activates *at all*, more often under forced than planned outages. Writing $p_{\blacksquare} = P(\text{IVA_it} > 0)$, the **logit model** relates the *log-odds* of activation linearly to the regressor:

$$H_{\blacksquare}: \gamma_{\blacksquare} = 0 \text{ vs. } H_{\blacksquare}: \gamma_{\blacksquare} > 0$$

Chapter 22. H6 in Full: The Placebo Test

Same specification family as H1–H4, but fit on a variable that is structural (Chapter 2) and should be inert to any single outage.

$$H_{\blacksquare}: \beta_{\blacksquare} = 0 \text{ vs. } H_{\blacksquare}: \beta_{\blacksquare} \neq 0 \text{ (the project WANTS to fail to reject } H_{\blacksquare} \text{ here)}$$

Chapter 23. Economic Significance: Thresholds and the Decision Table

A coefficient can be statistically significant — reliably distinguishable from zero, per Chapters 14–16 — and still be economically irrelevant: a rock-solid, $p < 0.0001$ shift of half a megawatt in a thousand-megawatt RAM budget is not something anyone would act on.

Chapter 24. Collinearity Between the Dummy and the Dose

Each outage type contributes both a binary "active" dummy and a continuous "MW lost" dose variable to the regressors Chapters 17–22 can use.

Chapter 25. From Raw Outage to Verdict: A Complete Worked Walkthrough

Tying Chapters 5–24 together, start to finish, for one hypothesis:

Part IV — Counterfactual Modeling: Foundations

Chapter 26. What “Counterfactual” Means, and the One Rule

Part III's regressions test whether outages *in general* move a parameter, pooling many events. “Single Event” asks a sharper question about *one specific outage*: what would this CNEC's series have looked like if the outage had never happened? Write the observed series as y_t , and let $t=0$ mark the outage's start.

$$impact_t = y_t - Y_{\square}(0)_t$$

Part V — The Method Catalog, Mathematically

Chapter 27. Deterministic Lookup: Seasonal-Naive and Hurdle

Seasonal-naive. The simplest possible $Y_{\blacksquare}(0)$: the pre-period average for each (hour, weekday) cell. Writing $h(t)$, $d(t)$ for the hour and weekday of timestamp t , and P for the set of pre-period timestamps:

$$Y_{\blacksquare}(0)_t = (\sum_{s \in P : h(s)=h(t), d(s)=d(t)} y_s) / |\{s \in P : h(s)=h(t), d(s)=d(t)\}|$$

Chapter 28. Harmonic Regression and STL Decomposition

Fourier + linear trend. A smooth parametric curve: a linear trend plus K pairs of sine/cosine harmonics at the data's natural period T , estimated by Chapter 5's OLS applied to this particular design matrix:

$$y_t = \beta_0 + \beta_1 t + \sum_{k=1}^K [a_k \sin(2\pi k t / T) + b_k \cos(2\pi k t / T)] + \varepsilon_t$$

Chapter 29. Box-Jenkins: ARIMA and SARIMA

Introduce the **backshift operator** B , defined by $B y_t = y_{t-1}$. An ARIMA(p, d, q) model says: after differencing the series d times to remove any trend ($\nabla^d y_t = (1-B)^d y_t$), the result follows an autoregressive-moving-average process,

$$\phi(B) (1-B)^d y_t = \theta(B) \varepsilon_t$$

Chapter 30. State-Space Models: Structural Time Series and TBATS

SARIMA models only *one* seasonal period at a time and bolts a second one on afterward via a separate lookup. **State-space (“structural”) models** instead represent the series as a sum of latent, separately-evolving components observed with noise:

$$y_t = \mu_t + \gamma_t + \varepsilon_t \text{ (observation equation)}$$

Chapter 31. The Theta Method

A top performer in the M3 and M4 forecasting competitions, specifically strong on *short* series — the regime where ARIMA's order search and the lag-feature methods of Chapter 33 both struggle for lack of data. The idea: decompose the (deseasonalised) series into two “theta-lines,” each a transform of the second differences $\nabla^2 y_t = y_t -$

$2y_{t-1} + y_{t-2}$ scaled by a parameter θ :

Theta-line(θ)_t has the same mean and slope as y_t , but its local curvature ∇^2 is scaled by θ

Chapter 32. Ridge Regression in Closed Form

Three methods (this one, LightGBM, and CatBoost in Chapter 33) share one feature set: for each timestamp, how much the target was 1, 2, and 7 days prior, plus their mean, plus cyclical hour/weekday indicators.

$$\beta_{ridge} = \operatorname{argmin}_{\beta} ||y - X\beta||^2 + \lambda ||\beta||^2$$

Chapter 33. Gradient-Boosted Trees: LightGBM and CatBoost

Where Ridge fits one global linear function, gradient boosting builds an *additive ensemble of decision trees*, each correcting the previous ensemble's mistakes.

$$r_{i,m} = y_i - F_{m-1}(x_i)$$

Chapter 34. Two Physics-Informed Methods, Mathematically

Some target columns are not independent physical quantities — they are *derived* from other columns already in the data, by a known formula. A model that respects that formula by construction should beat one that has to rediscover it statistically from a short pre-period.

$$Y_{(0)}^{RAM}_t = \sum_{k=1}^7 s_k \cdot X_{k,t}$$

Chapter 35. The Adaptive Ensemble, Mathematically

The ensemble backtests every *other* method on a held-out slice carved from the pre-period itself, computing each candidate method m's **mean absolute error**,

$$MAE_m = (1/n) \sum_{t \in \text{holdout}} |y_t - Y_{m,t}(0)|$$

Part VI — Touring the Tab, Sub-Tab by Sub-Tab

You now have both the vocabulary and the mathematics. This part walks the eight sub-tabs of Maintenance Analysis in the order a first-time user would actually click through them.

Chapter 36. Setup and Outage Sources

Setup is where a session begins: load whatever market data is already on the application's Fetch/Upload tab, or generate a synthetic ninety-day demonstration dataset on the spot. Setup carries its own UTC/CET timestamp-zone toggle, independent of every other tab's, so a different timestamp assumption can be tried for **this analysis** without disturbing anything already fetched elsewhere.

Chapter 37. Run Analysis and Reading Results

One button runs the full battery from Part III across **every** CNEC in the loaded dataset, never just the one selected elsewhere: entity fixed effects (Chapter 9) need at least two entities to be estimable at all. The selected CNEC is used only as a report label.

Chapter 38. Plots

Four chart types — a time series of the core parameters, a PTDF distribution, a shadow-price-versus-RAM scatter, and a CNEC binding-frequency bar chart — all drawn from the same population-level data Run Analysis just processed.

Chapter 39. Single Event: ITS, Δ RAM Decomposition, DiD, and the Per-CNEC Table

Pick one outage, a pre-period length, a post-event length, and one of the fifteen counterfactual methods from Part V — one function call then produces four views of the same event.

$$ATT = (high_{during} - high_{pre}) - (low_{during} - low_{pre})$$

Chapter 40. The Price Spread Pane, Mathematically

Isolates one CNEC's own change in contribution to a target zone's price. A CNEC's shadow price times a zone's PTDF sensitivity on that CNEC is that CNEC's own contribution to that zone's price; comparing the **actual** value against its **counterfactual** isolates the outage's own effect:

$$impact = (SP_{actual} \times PTDF_{actual}) - (SP_{cf} \times PTDF_{cf})$$

Chapter 41. Explain and Export

Explain turns the Results verdicts into a plain-language paragraph per hypothesis.

Export writes the covariates, outages, and verdicts to CSV, and can produce a self-contained HTML report embedding whatever charts are currently drawn.

Part VII — Maintaining This Tab

Chapter 42. State and Data Flow

Six pieces of state hold everything this tab knows, and every downstream sub-tab reads from these and nothing else.

Chapter 43. How This Differs From the Main Application

This tab and the main six-tab dashboard both ultimately call the same underlying analytical functions and should agree on every number for identical input — but the two paths to get there are **duplicated code, not shared code**. A fix made in one path does not automatically fix the other.

Chapter 44. Known Gotchas

Chapter 45. A Maintenance Checklist

When something in this tab looks wrong, check in this order:

Part VIII — Hypothesis Testing, Deepened

Parts II and III built the machinery and applied it once to each hypothesis. This part goes back and finishes the job properly: the asymptotic theory that justifies treating a t-statistic as approximately normal in the first place, the general Wald-test framework every single-coefficient test in this book is secretly an instance of, worked numeric examples for the three hypotheses (H2, H4, H6) Part III stated but did not fully compute by hand, and a side-by-side comparison of all six verdicts on one consolidated report.

Chapter 46. Asymptotic Properties: Consistency and Asymptotic Normality

Chapter 7's Gauss–Markov theorem is an exact, finite-sample result — it holds for any n , however small — but it required homoskedastic, independent errors, assumptions Chapters 9 and 13 already showed this data violates.

Chapter 47. The Wald Test: The General Framework Behind Every t-Test in This Book

Every single-coefficient test in Chapters 17–22 — "is $\beta_{\text{■}}$ different from zero" — is a special case of a more general tool. The **Wald test** for a linear restriction $R\theta=r$ on a parameter vector θ with estimated covariance $\text{Var}(\theta_{\text{■}})$ is

$$W = (R\theta_{\text{■}} - r)_{\text{■}} [R \cdot \text{Var}(\theta_{\text{■}}) \cdot R_{\text{■}}]^{-1} (R\theta_{\text{■}} - r)$$

Chapter 48. H2, Worked: AC Outages and PTDF Sensitivity, With Real Numbers

Recall H2's specification from Chapter 18: $|\text{PTDF_SRC}|_{\text{it}} = \alpha_{\text{it}} + \beta_{\text{■}} \cdot \text{ACOutage}_{\text{it}} + \varepsilon_{\text{it}}$. Two CNECs, six observations each, three before an AC-line outage and three during it:

$$\beta_{\text{■}} = 0.00983, SE = 0.00098, t = 10.00, p < 0.0001 \text{ (dof} = 12 - 2 - 1 = 9\text{)}$$

Chapter 49. H4, Worked: Forced Outages and the Shadow Price, With Real Numbers

Recall H4's specification from Chapter 20, fit only on binding rows ($\text{shadowPrice} > 0$). Two CNECs, four binding observations each, two pre- and two during a forced outage:

$$\beta_{\text{H6}} = 11.75, SE = 1.328, t = 8.85, p = 0.0003 \text{ (dof} = 8-2-1 = 5\text{)}$$

Chapter 50. H6, Worked: The Placebo, Confirmed Null

Recall H6's specification from Chapter 22 — the one hypothesis this project *wants* to fail to reject. Two CNECs, six observations each, three before and three during an outage, on FRM specifically:

$$\beta_{\text{H6}} = -0.033, SE = 0.113, t = -0.294, p = 0.775 \text{ (dof} = 12-2-1 = 9\text{)}$$

Chapter 51. Comparing All Six Verdicts Side by Side

Consolidating this part's and Part III's worked numbers into one report-style table — the shape `summarize_hypotheses()` actually produces, read all at once rather than one hypothesis at a time:

Part IX — Diagnostics and Robustness

Every regression in Part III rests on assumptions — Chapter 4's four, plus homoskedasticity for the classical formulas, relaxed piece by piece through Part II. This part covers how to check whether those assumptions actually held on a given fit, and what to do when they visibly didn't.

Chapter 52. Checking the Residuals: What a Well-Behaved Fit Looks Like

A regression's residuals ($\hat{u}_i = y_i - \hat{y}_i$) are the closest thing available to the unobservable ϵ_i , and most of Chapter 4's assumptions have a direct residual-based check:

Chapter 53. Autocorrelation and the Durbin–Watson Statistic

Chapter 13 addressed correlation *across entities at the same timestamp* via clustering. A related but distinct concern is correlation *within one entity's own series over time* — today's residual predicting tomorrow's, which clustered-by-time standard errors do not, by themselves, detect or fix.

$$DW = \frac{\sum_{t=2}^n (\hat{u}_t - \hat{u}_{t-1})^2}{\sum_{t=1}^n \hat{u}_t^2}$$

Chapter 54. Sensitivity to Pre-Period Length and Clustering Choice

Two choices a user makes on the Setup and Single Event sub-tabs (Chapters 36, 39) have a direct line back to this part's theory, and are worth checking deliberately rather than left at their defaults:

Chapter 55. A Robustness Checklist

Before trusting a SUPPORTED verdict from this tab, or reporting one further:

Part X — A Complete Worked Case Study

Chapter 56. From Raw Outage Event to Every One of Six Verdicts: A Full Numeric Walkthrough

Tying every chapter in Parts II, III, and VIII together, start to finish, on one hypothetical but internally consistent scenario: a forced HVDC outage between a source country and two of a target zone's CNECs (A and B), covering three MTUs before the outage and three during it in the illustrative panel already used throughout Chapters 6, 10, 17–22, and 48–50.

Appendix A — Notation Reference

Symbol	Meaning
i	Entity index (a CNEC)
t	Time index (an MTU / timestamp)
g	Cluster index (Chapter 13)
y_{it}, x_{it}, D_{it}	Outcome, generic regressor, treatment/outage dummy for entity i at time t
$\alpha, \beta, \gamma, \theta, \phi$	Population parameters (intercept, slope(s), logit coefficients, ARIMA/Theta parameters)
α_i	Entity fixed effect (Chapter 9)
ε_{it}, u_{it}	Error term / residual
hat, e.g. $\hat{\beta}$	An estimated (sample) quantity
bar, e.g. $\bar{\beta}$	A sample average
tilde, e.g. $\tilde{\beta}$	A within-transformed (demeaned) quantity
X, y	Design matrix, outcome vector (matrix notation)
$SE(\cdot)$	Standard error of an estimator
n, m, G, k	Sample size; number of hypotheses/harmonics; number of clusters; number of parameters
σ^2	Error variance
λ	Ridge penalty strength (Chapter 32)
η	Gradient-boosting learning rate (Chapter 33)
B	Time-series backshift operator, $B y_t = y_{t-1}$ (Chapter 29)
$Y_{\square(0)_t}$	Counterfactual projection at time t (Chapter 26)
s_k	Sign of the k -th RAM-identity term (Chapter 3)

Appendix B — Formula Quick Reference

Estimation (Part II)

$$\beta = (X'X)^{-1}X'y \text{ [OLS, Ch. 5]}$$

$$y_{it} = y_{it} - \alpha_i \text{ [within transform, Ch. 9]}$$

$$\text{Var_cluster}(\beta) = (X'X)^{-1}[\sum_g X_g u_{gu} X_g](X'X)^{-1} \text{ [Ch. 13]}$$

$$t = \beta/SE(\beta), p_{-}(k) \leq \alpha/(m-k+1) \text{ [Ch. 14, 16]}$$

The Six Hypotheses (Part III)

$$H1: fall_{it} = \alpha_i + \beta \cdot HVDCOutage_{it} + \epsilon_{it}$$

$$H2: |PTDF_SRC|_{it} = \alpha_i + \beta \cdot ACOutage_{it} + \epsilon_{it}$$

$$H3: RAM_{it} = \alpha_i + \beta \cdot ForcedOutage_{it} + \epsilon_{it}$$

$$H4: shadowPrice_{it} = \alpha_i + \beta \cdot ForcedOutage_{it} + \epsilon_{it} \text{ (binding rows only)}$$

$$H5: \ln(p_{it}/(1-p_{it})) = \gamma + \gamma \cdot Forced_{it}$$

$$H6: FRM_{it} = \alpha_i + \beta \cdot Outage_{it} + \epsilon_{it} \text{ (placebo)}$$

Counterfactual Methods (Part VI)

$$RAM = \sum_k s_k X_k \text{ [Ch. 3, 34]}$$

$$impact = (SP_a \times PTDF_a) - (SP_{cf} \times PTDF_{cf}) \text{ [Ch. 40]}$$

Appendix C — Glossary

Statistics and Econometrics

Term	Definition
AIC	Akaike Information Criterion, $-2\ln(L) + 2k$ — a fit-vs-complexity model-selection score (Ch. 29)
Asymptotic	Describing a property that holds as sample size grows without bound, not necessarily at any finite n (Ch. 46)
BLUE	Best Linear Unbiased Estimator (Gauss–Markov theorem, Ch. 7)
Confidence interval	A range of parameter values consistent with the data at a stated confidence level (Ch. 14)
Consistency	An estimator converging in probability to the true parameter as $n \rightarrow \infty$ (Ch. 46)
Degrees of freedom (df)	Roughly, the number of independent pieces of information left after estimating a model's parameters; divides SSE to get an unbiased σ^2 (Ch. 6)
Durbin–Watson statistic	A test for first-order autocorrelation in regression residuals (Ch. 53)
Fixed effects	Entity-specific intercepts that absorb time-invariant differences between panel members (Ch. 9)
FWER	Family-wise error rate — probability of at least one false positive across multiple tests (Ch. 16)
FWL theorem	Frisch–Waugh–Lovell — demeaning/residualising equals “controlling for” (Ch. 11)
Heteroskedasticity	Error variance that differs across observations (Ch. 12)
Holm–Bonferroni	A step-down multiple-testing correction, more powerful than plain Bonferroni (Ch. 16)
LSDV	Least Squares Dummy Variables — the (equivalent, costlier) alternative to the within estimator (Ch. 9)
MLE	Maximum-likelihood estimation — choosing parameters to maximise the probability of the observed data (Ch. 21)
Normal equations	The system $\partial S / \partial \beta = 0$ whose solution is the OLS estimator (Ch. 5)
p-value	The probability, under the null hypothesis, of a test statistic at least as extreme as the one observed (Ch. 14)
Panel data	Repeated observations of the same cross-sectional units over time (Ch. 9)

Term	Definition
Power	Probability of correctly detecting a real effect (Ch. 15)
Robust (White) SE	A standard error valid under heteroskedasticity of unknown form (Ch. 12)
Sandwich estimator	A variance formula of the form $(X'X)^{-1}[\dots](X'X)^{-1}$, used for both robust and clustered SEs (Ch. 12, 13)
t-statistic	$\beta_j/SE(\beta_j)$, the standardised distance of an estimate from a hypothesised value (Ch. 14)
Type I / II error	False positive / false negative (Ch. 15)
Wald test	The general test-statistic framework a single-coefficient t-test is a special case of (Ch. 47)
Within estimator	Fixed-effects regression via demeaning (Ch. 9)

Time Series and Machine Learning

Term	Definition
ARIMA	AutoRegressive Integrated Moving Average — a Box-Jenkins time-series model (Ch. 29)
Backshift operator (B)	$B y_t = y_{t-1}$, notation for writing time-series models compactly (Ch. 29)
Box-Cox transform	A power transform y^λ used to stabilise variance before modeling (Ch. 30)
Ensemble	A blend of several models' predictions, here weighted by inverse-squared backtest error (Ch. 35)
Gradient boosting	Building an ensemble of trees, each fit to the previous ensemble's residual/gradient (Ch. 33)
Kalman filter	A recursive predict-correct algorithm for estimating a state-space model's latent states (Ch. 30)
Learning rate (η)	The shrinkage factor applied to each new tree in gradient boosting (Ch. 33)
LOESS	Locally weighted regression, the smoothing technique behind STL (Ch. 28)
Pearson correlation (r)	A standardised measure of linear association between two series (Ch. 34)
Ridge regression	Penalised least squares, $(X'X + \lambda I)^{-1}X'y$, for collinear features (Ch. 32)
SARIMA	Seasonal ARIMA — ARIMA with an added seasonal backshift polynomial (Ch. 29)

Term	Definition
State-space model	A model representing a series as latent, evolving components observed with noise (Ch. 30)
TBATS	Box-Cox, ARMA errors, Trend, multiple Seasonal components — a state-space variant for complex seasonality (Ch. 30)
Theta method	A short-series forecasting method blending a trend line and a curvature-scaled SES line (Ch. 31)

Domain (FBMC)

Term	Definition
AMR	Adjustment for Minimum RAM (Ch. 2)
CNEC	Critical Network Element with Contingency (Ch. 2)
FAAC / AAC	Already Allocated Capacity (Ch. 2)
fall / F_allReference	The CNEC reference flow entering the RAM formula (Ch. 2)
FBMC	Flow-Based Market Coupling (Ch. 1)
fnrao / FRA	Flow from Non-costly Remedial Actions and Other adjustments (Ch. 2)
FRM	Flow Reliability Margin (Ch. 2)
IVA	Individual Validation Adjustment (Ch. 2)
MTU	Market Time Unit, 15 minutes (Ch. 2)
PTDF	Power Transfer Distribution Factor (Ch. 2)
RAM	Remaining Available Margin (Ch. 3)
Shadow price	The marginal value of relaxing a binding CNEC constraint (Ch. 2)

Appendix D — Further Reading

This book derives everything it uses, but none of it was invented here. For a reader who wants the primary or textbook sources behind each tool:

Topic	Where to go deeper
OLS, fixed effects, clustering	Wooldridge, <i>*Introductory Econometrics*</i> (panel-data chapters); Angrist & Pischke, <i>*Mostly Harmless Econometrics*</i>
Gauss–Markov, asymptotics, the Wald test	Greene, <i>*Econometric Analysis*</i>
Multiple testing (Holm–Bonferroni)	Holm (1979), “A Simple Sequentially Rejective Multiple Test Procedure,” <i>*Scandinavian Journal of Statistics*</i>
ARIMA / SARIMA (Box–Jenkins)	Box, Jenkins, Reinsel & Ljung, <i>*Time Series Analysis: Forecasting and Control*</i>
STL decomposition	Cleveland, Cleveland, McRae & Terpenning (1990), “STL: A Seasonal-Trend Decomposition Procedure Based on Loess”
State-space models and the Kalman filter	Durbin & Koopman, <i>*Time Series Analysis by State Space Methods*</i>
TBATS	De Livera, Hyndman & Snyder (2011), “Forecasting Time Series With Complex Seasonal Patterns Using Exponential Smoothing”
The Theta method	Assimakopoulos & Nikolopoulos (2000), “The Theta Model: A Decomposition Approach to Forecasting”
Ridge regression	Hoerl & Kennard (1970), “Ridge Regression: Biased Estimation for Nonorthogonal Problems”
Gradient boosting	Friedman (2001), “Greedy Function Approximation: A Gradient Boosting Machine”
FBMC methodology	The relevant TSOs' own Nordic (or Core) Flow-Based Market Coupling methodology documentation, published by the region's NEMOs/TSOs

— End —